

title: "Savitribai Phule Pune University --- Programming and Problem Solving"
code: "110005"
pattern: "2019"
total_marks: 70
duration: "2½ Hours"

Seat No.

SAVITRIBAI PHULE PUNE UNIVERSITY

Programming and Problem Solving (110005)

2019 Pattern | Semester I/II

Total Marks: 70 | Time: 2½ Hours

Instructions: Q1/Q2, Q3/Q4, Q5/Q6, Q7/Q8. Neat diagrams must be drawn wherever necessary.

Unit III --- Functions and Modules (18 Marks)

Q1) Attempt the following: [15]

a) Define a function. Explain the need for functions in Python. [6]

b) Differentiate between local and global variables with examples. [4]

c) Write a program using a function to find the largest of 3 numbers by passing numbers as arguments. [5]

OR

****Q2)** Attempt the following:** [15]

a) Explain the following types of function arguments with examples: i) **Default arguments** [6]
ii) **Variable length arguments**

b) Write a program to find the square of a number using lambda function. [5]

c) What are modules and packages in Python? Explain a user-defined module with example. [4]

Unit IV --- Strings (17 Marks)

Q3) Attempt the following: [15]

a) Explain different ways of formatting a string in Python with suitable examples. [6]

b) Explain different ways of iterating a string. [4]

c) What is the output of the following for string `S = "Learning Python is fun": i)`

`print(S[:10])`

ii) `print(S[::-1])` iii) `print("Is" not in S)` iv) `print(S<span class="marks">[4]`

`</span>)` v)

`print(S[0:10])`

[5]

OR

****Q4)** Attempt the following:** [15]

a) Explain the following string methods with examples: i) `isspace()` ii) `join()` [6]

iii) `lower()`

- b) Write a program to display a string and count the number of characters in the string using a loop. [5]
- c) Explain the `ord()` and `chr()` functions with examples. Explain `in` and `not in` operators on strings. [4]
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Unit V --- Object Oriented Programming (18 Marks)

Q5) Attempt the following: [15]

- a) Explain the following programming paradigms: i) Monolithic Programming ii) Procedural Programming iii) Object Oriented Programming [6]
- b) Explain the following OOP concepts with examples: i) Class variable ii) Object variable [5]
- c) Write a Python program to create a class `Employee` with attributes `Name`, `emp_id`, and `salary`. Display data of 4 employees. [4]

OR

****Q6)** Attempt the following:** [15]

- a) Explain the following features with examples: i) Containership ii) Inheritance iii) Data Abstraction [6]
- b) Explain class method and self object with suitable example. [5]
- c) Write a program to create a class `Car` with two attributes `name` and `cost`. Create two objects and display their information. [4]
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Unit VI --- File Handling (17 Marks)

Q7) Attempt the following: [15]

- a) What is a file? Explain relative and absolute path of a file with examples. [6]
- b) Explain the following file handling methods: i) `read(n)` ii) `readlines()` iii) `close()` [5]
- c) Explain the following file access modes: `r`, `w+`, `a+`, `rb`, `ab+`. [4]

OR

****Q8)** Attempt the following:** [15]

- a) Explain any three directory methods with examples. [6]
- b) Explain text and binary files with examples. [5]
- c) Explain any three dictionary methods with examples. [4]
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*** END OF QUESTION PAPER ***